

$$\text{Form A} \quad \bar{a} = 1 - e^{-b(\bar{w} - \bar{w}_s)} \quad p = 0$$

$$\text{Form B} \quad \bar{a} = 1 - \frac{b}{(\bar{w} - \bar{w}_s) + b} \quad p = 1$$

$$\text{Form C} \quad \bar{a} = 1 - \frac{b}{1 + \bar{w} - \bar{w}_s} \quad p = 1$$

$$\bar{w}_s \text{ known,} \quad \bar{a}'_{\text{true}} := \frac{d\bar{a}_{\text{true}}}{d\bar{w}}$$

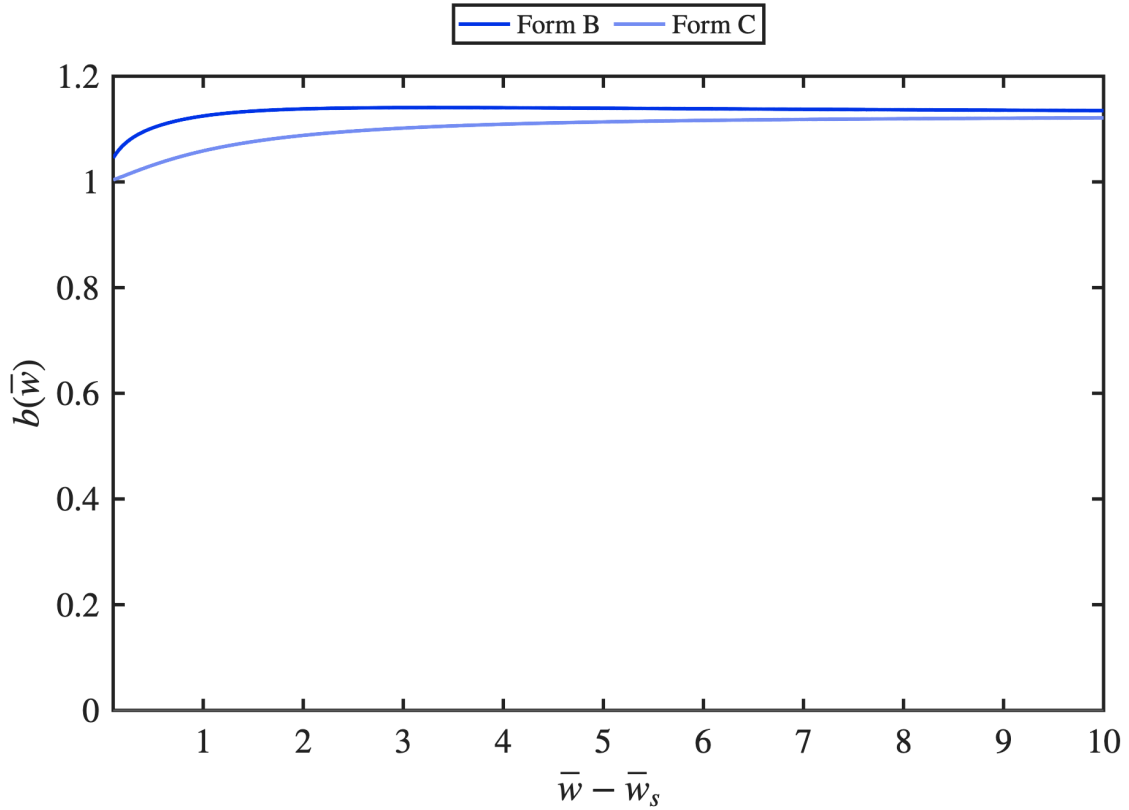
$$b_A(\bar{w}) = -\frac{\ln(1 - \bar{a}_{\text{true}})}{\bar{w} - \bar{w}_s}, \quad b_B(\bar{w}) = (\bar{w} - \bar{w}_s) \frac{1 - \bar{a}_{\text{true}}}{\bar{a}_{\text{true}}}, \quad b_C(\bar{w}) = (1 + \bar{w} - \bar{w}_s)(1 - \bar{a}_{\text{true}})$$

$$\bar{w} - \bar{w}_s \in [0.1, 10]$$

$$\bar{w} - \bar{w}_s = 0 : \quad \text{A, B : any } b ; \quad \text{C : } b = 1$$

$$\bar{w} - \bar{w}_s = \infty : \quad \text{A, B, C : any } b$$

$$\text{elsewhere : } \partial \bar{a} / \partial b \neq 0 \Rightarrow b(\bar{w}) \text{ unique}$$



$$\bar{a}'(\bar{w}; b) := \left. \frac{\partial \bar{a}}{\partial \bar{w}} \right|_b \quad \text{evaluated at} \quad b = b(\bar{w})$$

$$\bar{a}(\bar{w}; b(\bar{w})) - \bar{a}_{\text{true}} = 0 \quad (1.1 \times 10^{-16})$$

$$\Delta \bar{a}' := \bar{a}'(\bar{w}; b(\bar{w})) - \bar{a}'_{\text{true}} = -\frac{\partial \bar{a}}{\partial b} \frac{db(\bar{w})}{d\bar{w}} \quad (1.5 \times 10^{-16})$$

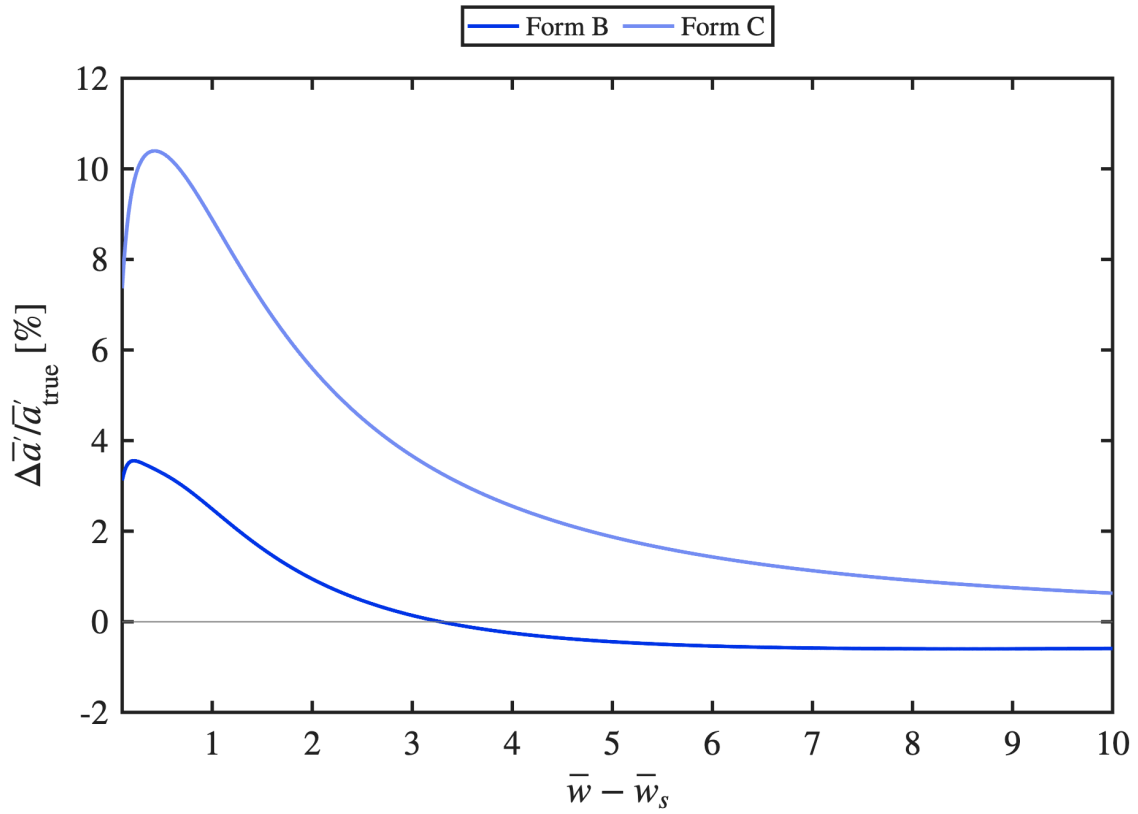
$$\Delta \bar{a}' / \bar{a}'_{\text{true}} \text{ [%] } :$$

$$\text{RMS : } 106.5, \quad 1.19, \quad 4.26 \text{ \%}$$

$$\text{sup : } 152.8, \quad 3.56, \quad 10.40 \text{ \%}$$

$$\text{range : } [+7.9, +152.8], \quad [-0.60, +3.56], \quad [+0.63, +10.40] \text{ \%}$$

$$\text{zero at } \bar{w} - \bar{w}_s : \quad \text{none}, \quad 3.284, \quad \text{none}$$

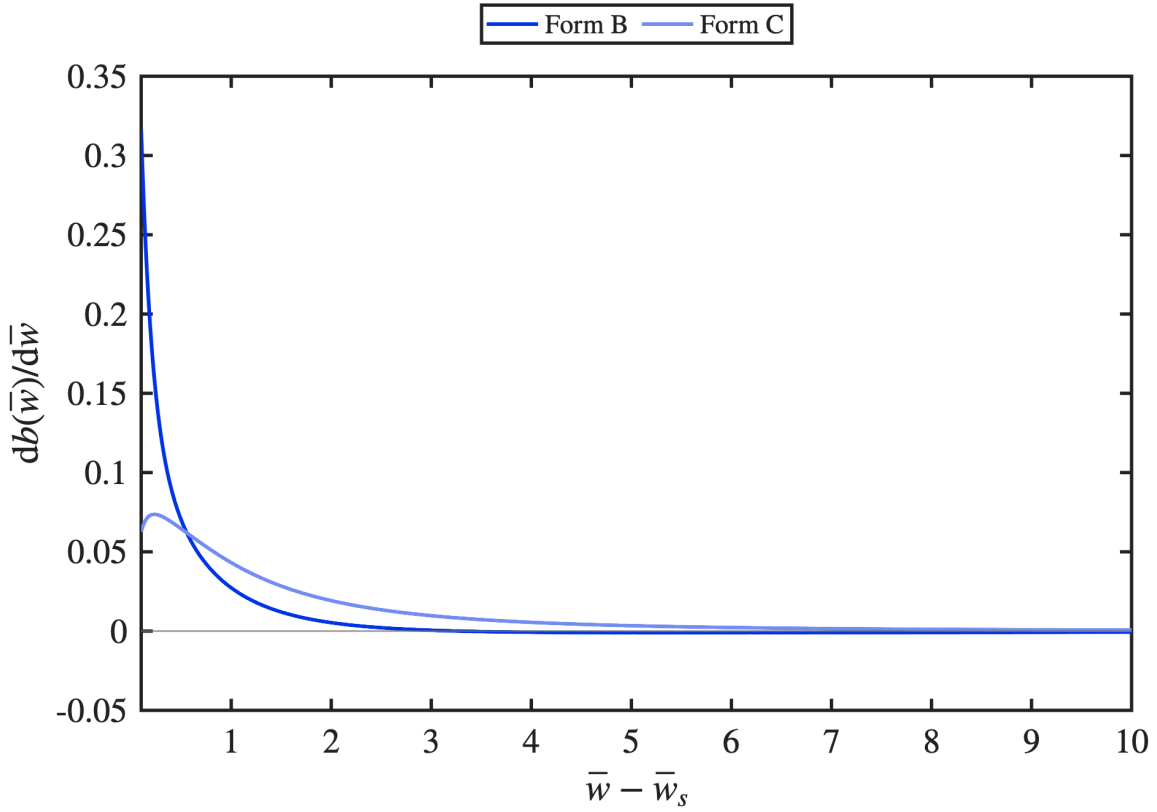


$$\frac{db_A(\bar{w})}{d\bar{w}} = \frac{1}{(\bar{w} - \bar{w}_s)^2} \left[\frac{(\bar{w} - \bar{w}_s) \bar{a}'_{\text{true}}}{1 - \bar{a}_{\text{true}}} + \ln(1 - \bar{a}_{\text{true}}) \right]$$

$$\frac{db_B(\bar{w})}{d\bar{w}} = \frac{1 - \bar{a}_{\text{true}}}{\bar{a}_{\text{true}}} - (\bar{w} - \bar{w}_s) \frac{\bar{a}'_{\text{true}}}{\bar{a}_{\text{true}}^2}$$

$$\frac{db_C(\bar{w})}{d\bar{w}} = (1 - \bar{a}_{\text{true}}) - (1 + \bar{w} - \bar{w}_s) \bar{a}'_{\text{true}}$$

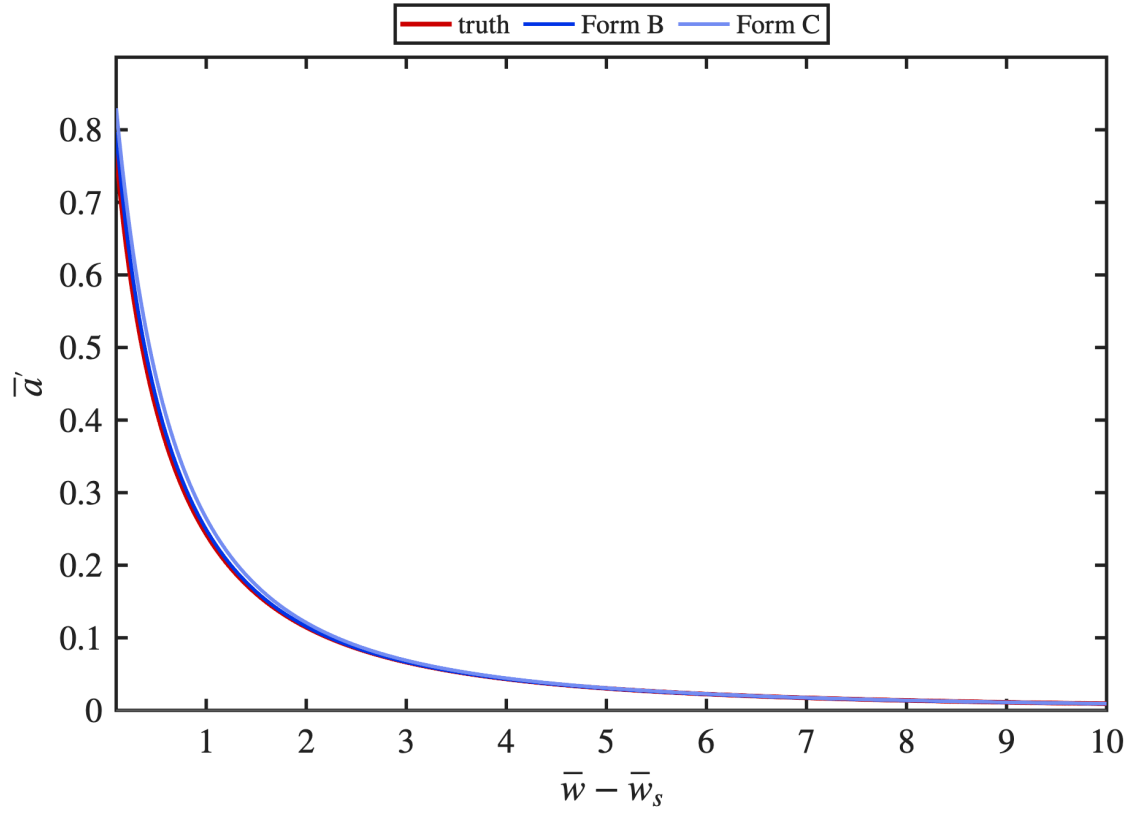
$$\left. \frac{db(\bar{w})}{d\bar{w}} \right|_{\bar{w} - \bar{w}_s = 0.1} = -0.669, +0.316, +0.063 \quad \left. \frac{db(\bar{w})}{d\bar{w}} \right|_{\bar{w} - \bar{w}_s = 10} = -0.014, -0.001, +0.001$$



$$\bar{a}'_{\text{A}} = -\frac{(1 - \bar{a}_{\text{true}}) \ln(1 - \bar{a}_{\text{true}})}{\bar{w} - \bar{w}_s}$$

$$\bar{a}'_{\text{B}} = \frac{\bar{a}_{\text{true}}(1 - \bar{a}_{\text{true}})}{\bar{w} - \bar{w}_s}$$

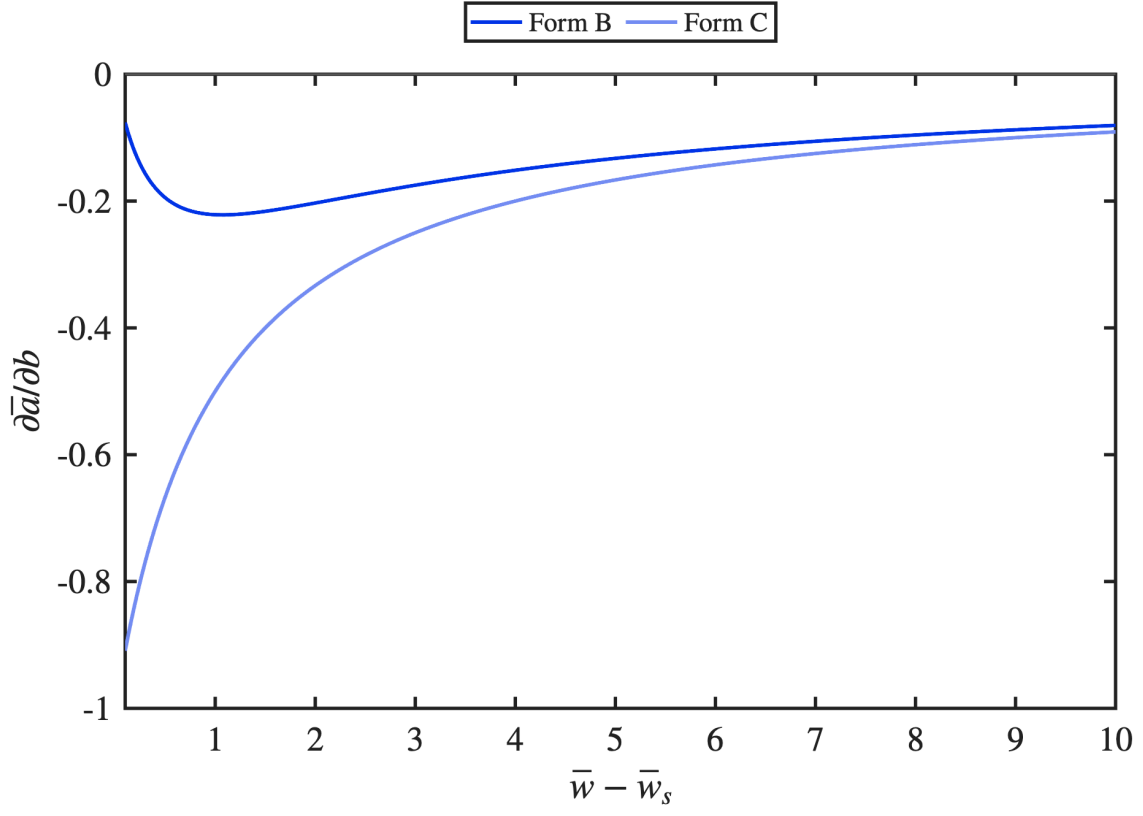
$$\bar{a}'_{\text{C}} = \frac{1 - \bar{a}_{\text{true}}}{1 + \bar{w} - \bar{w}_s}$$



$$\frac{\partial \bar{a}_A}{\partial b} = +(\bar{w} - \bar{w}_s)(1 - \bar{a}_{\text{true}})$$

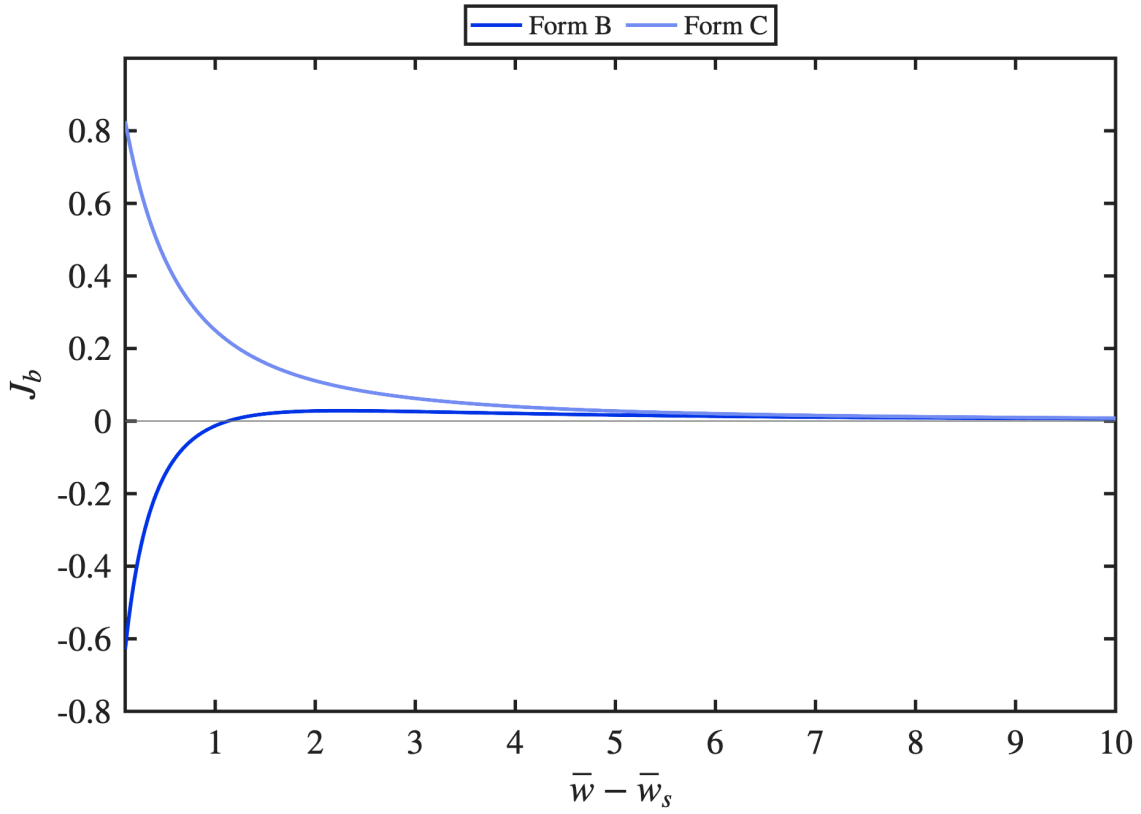
$$\frac{\partial \bar{a}_B}{\partial b} = -\frac{\bar{a}_{\text{true}}^2}{\bar{w} - \bar{w}_s}$$

$$\frac{\partial \bar{a}_C}{\partial b} = -\frac{1}{1 + \bar{w} - \bar{w}_s}$$



$$J_{b,A} = (1 - \bar{a}_{\text{true}}) \left[1 + \ln(1 - \bar{a}_{\text{true}}) \right] \quad J_{b,B} = \frac{\bar{a}_{\text{true}}^2 (2\bar{a}_{\text{true}} - 1)}{(\bar{w} - \bar{w}_s)^2} \quad J_{b,C} = \frac{1}{(1 + \bar{w} - \bar{w}_s)^2}$$

$$J_b = 0 : \quad \text{A : } \bar{a}_{\text{true}} = 1 - e^{-1}, \quad \bar{w} - \bar{w}_s = 1.955 \quad \text{B : } \bar{a}_{\text{true}} = \frac{1}{2}, \quad \bar{w} - \bar{w}_s = 1.128 \quad \text{C : none}$$



$$\bar{a}_{\text{true}} = \frac{1}{c}$$

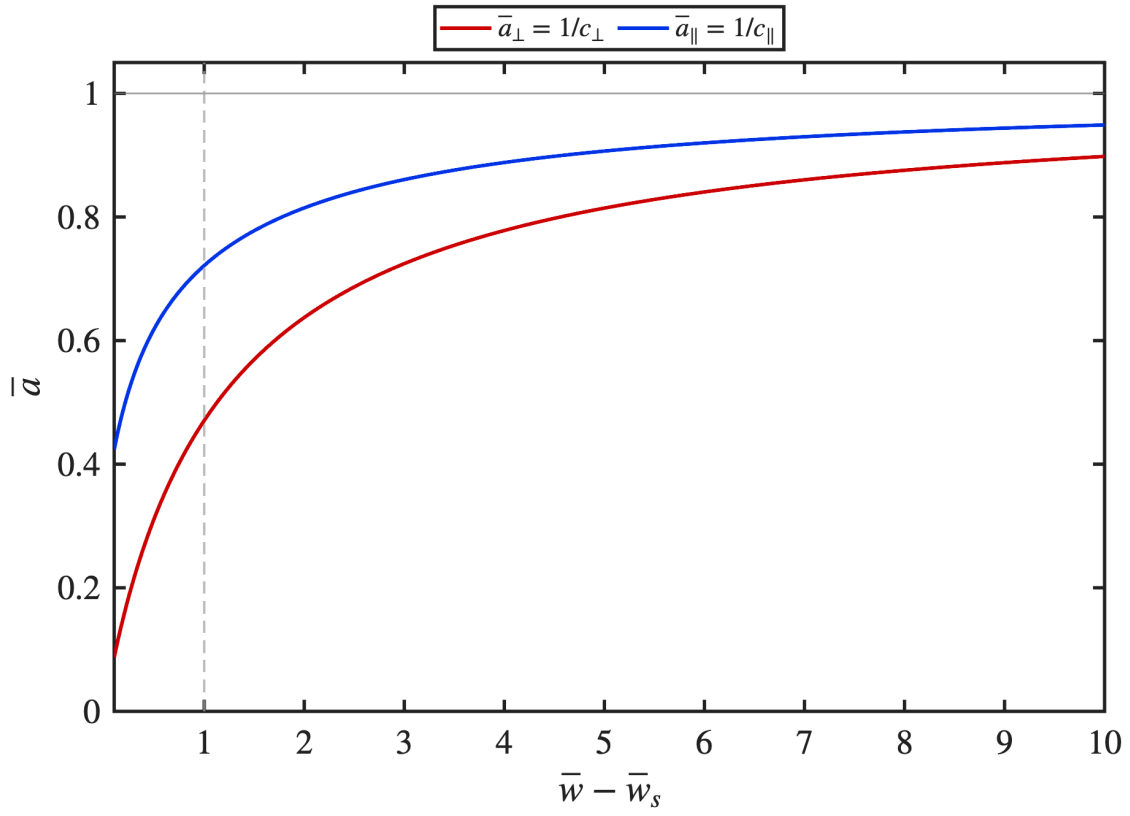
$c_{\perp}, c_{\parallel}$ two-sphere series

$$\bar{w}_s = 1$$

$$\bar{w} - \bar{w}_s = 1 : \quad \bar{a}_{\perp} = 0.4706, \quad \bar{a}_{\parallel} = 0.7214$$

$$\bar{w} - \bar{w}_s = 10 : \quad \bar{a}_{\perp} = 0.8981, \quad \bar{a}_{\parallel} = 0.9489$$

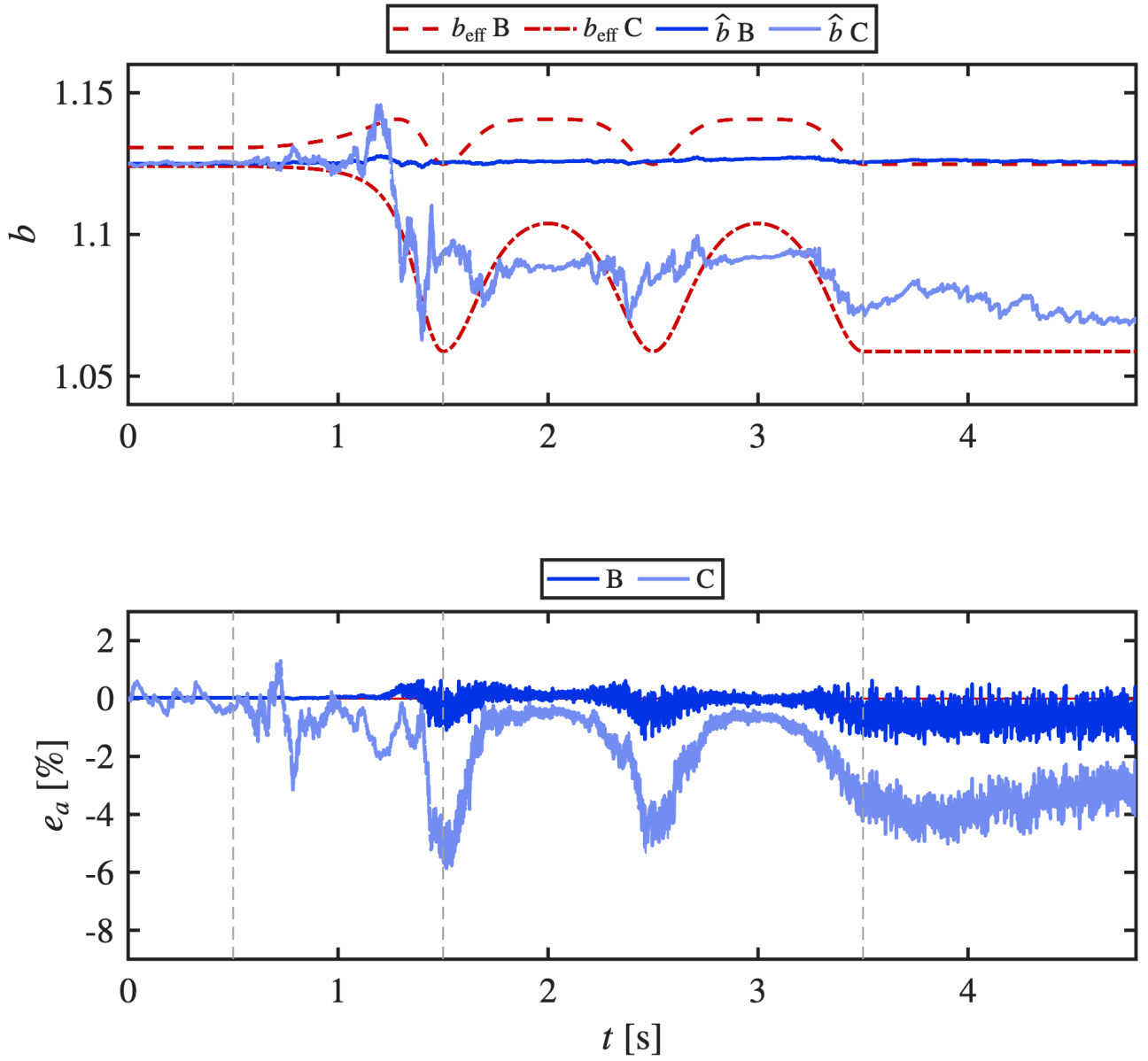
$$\bar{w} - \bar{w}_s \rightarrow \infty : \quad (1 - \bar{a}_{\perp})\bar{w} \rightarrow \frac{9}{8}, \quad (1 - \bar{a}_{\parallel})\bar{w} \rightarrow \frac{9}{16}$$



truth = $c_\perp$
 $\hat{b}$ estimated

$b_0 = \frac{9}{8}$ (both)
 $\sqrt{P[0]} = 0.0157$ (B), 0.0708 (C)

$p = 1, \bar{w}_s = 1$
 one seed



truth = $c_{\parallel}$ $b_0 = \frac{9}{8}$ (both, as on p.7) $p = 1, \bar{w}_s = 1$
 $\hat{b}$ estimated $\sqrt{P[0]} = 0.7505$ (B), 0.5697 (C) one seed

$c_{\parallel}$ substituted into the wall-normal channel; anchor kept, $\sqrt{P[0]} = \sup |b_{\text{eff}} - \frac{9}{8}|_{c_{\parallel}}$

